FIT2179 DV2

Why

This project focuses on Australia's domestic aviation network because it provides a strong case for data-driven visual storytelling at both national and regional scales. Air travel is a critical part of Australian mobility due to long distances, uneven population distribution, and the strong economic and social links between capital cities and regional centres. However, the importance of aviation is not experienced uniformly across the country. Some routes and airports dominate total passenger activity, while other places become more significant when traffic is considered relative to population, remoteness, or operational intensity. This makes the topic analytically rich: it supports a narrative that moves from large-scale concentration to disruption and recovery, and finally to more localised questions of dependence and hierarchy.

What

The webpage examines how Australia's domestic aviation system is shaped by concentration, disruption, hierarchy, and regional dependence. The first section establishes the spatial structure of the network by showing that a small number of major corridors, particularly those linked to Sydney, Melbourne, and Brisbane, account for a disproportionate share of passenger traffic. The second section focuses on temporal change, using time-series charts to show the sharp contraction during the COVID disruption period and the later recovery of both passenger volumes and load factors. The third section shifts to airport hierarchy, showing that the dominance of a few major hubs is not only visible in one month of traffic, but remains evident across longer time scales. The fourth section introduces alternative denominators and operational measures, including passengers per resident, aircraft movements, and passengers per movement, to show that the largest airports are not always the most revealing when interpreting dependence or intensity. The final seasonal view adds another layer by showing that monthly traffic patterns differ across major airports and are not equally stable across the network.

Who

The intended audience is a general Australian reader with an interest in transport, geography, infrastructure, or public systems, but without specialist knowledge of aviation data. The page is therefore designed to be explanatory rather than technical. It avoids assuming prior familiarity with metrics such as load factor or airport efficiency, and instead uses short interpretive notes, grouped visuals, and section-based storytelling to guide the reader through increasingly specific levels of analysis. The goal is to make the structure of the network legible to non-expert readers while still offering enough analytical depth to reward closer inspection.

How

The project combines multiple real-world datasets. The core aviation data is drawn from BITRE domestic airline activity, route-level time-series data, and airport traffic datasets, while ABS population context is used to support urban-area and per-resident comparisons. The webpage is built as a single publicly accessible GitHub Pages site, and all assessed visualisations are implemented in Vega-Lite. A range of visual forms is used to suit different analytical purposes, including maps, ranking charts, time-series charts, small multiples, comparison charts, a bump chart, a scatter plot, and a heatmap. The design emphasises narrative flow, visual hierarchy, and traceability through linked data files and Vega-Lite specifications, so that the project functions not only as a story page but also as a transparent and inspectable analytical submission.